

SIH26138: Quantum-Inspired Optimization Benchmarking & Validation

Executive Decision Report | Phase 3 Roadmap Deliverable | Generated: 2026-09-27 00:42:49 UTC

1. Objective-Specific Metric Leaders & Algorithmic Specialization

Operational Dimension	Leading Algorithm	Key Computational Advantage	Status
Lowest Fuel Consumption	Quantum-Inspired PSO (QPSO)	Hydrodynamic wave resistance tunneling	VERIFIED
Lowest Operational Cost	Quantum-Inspired PSO (QPSO)	Multi-tier capacity and speed co-optimization	OPTIMAL
Lowest GHG Emissions	Quantum-Inspired PSO (QPSO)	Recycle WTW alternative fuel compliance	IMO 2030
Highest Reliability	Quantum-Inspired PSO (QPSO)	Buffer-conscious arrival schedule guarantee	SATISFIED
Highest Demand Satisfaction	Quantum-Inspired PSO (QPSO)	Cargo delivery fulfillment guarantee	DELIVERED
Fastest Runtime Latency	Greedy Allocation	Direct non-iterative heuristic dispatch	REAL-TIME
Best Composite Objective	Quantum-Inspired PSO (QPSO)	Global delta-potential fitness dominance	QUANTUM LEAD

2. Canonical Algorithm Comparison Matrix (Base Scenario: 250,000t Demand)

Algorithm	Time (s)	Iter	Fuel (t)	Cost (\$M)	Emissions (t)	Reliability	Demand	Status
Quantum-Inspired PSO (QPSO)	0.394	50	221.4	\$2.71	851.5	100.0%	100%	FEASIBLE
Classical PSO	0.368	50	221.4	\$2.71	851.5	100.0%	100%	FEASIBLE
Genetic Algorithm	0.573	50	223.3	\$2.71	858.8	100.0%	100%	FEASIBLE
Simulated Annealing	0.032	50	15,286.1	\$151.04	48,574.8	100.0%	100%	INFEASIBLE
Linear Programming (Relaxed)	0.007	1	474.2	\$5.45	1,823.5	100.0%	100%	FEASIBLE
Greedy Allocation	0.000	1	504.5	\$5.48	1,939.8	100.0%	100%	FEASIBLE

3. Stochastic Stability & Monte Carlo Repeatability (N=30 Independent Seeds)

Algorithm	Runs	Mean Objective	Std Dev	Best Score	Worst Score	CV (%)	Consistency
QPSO	30	7.8	0.00	7.8	7.8	0.01%	HIGH (Stable)
Classical PSO	30	9.3	3.17	7.8	15.7	33.89%	MODERATE
Genetic Algorithm	30	14.6	9.13	7.8	37.0	62.54%	MODERATE
Simulated Annealing	30	410.0	276.27	16.8	1,106.1	67.38%	MODERATE

4. Hydrodynamic Fuel Consumption Prediction Benchmarking

Model Architecture	MAE (t)	RMSE (t)	R² Score	Bias (t)	Residual Std	Latency (ms)
Linear Regression (Baseline)	3.365	5.867	0.5665	-0.273	5.861	2.50 ms
Random Forest	0.754	1.681	0.9644	+0.035	1.681	99.98 ms
Histgbdt (Production Tree)	0.743	1.549	0.9698	-0.006	1.549	26.54 ms
Quantum-Inspired Predictor (Qifcp)	3.700	5.604	0.6044	-0.204	5.601	2.52 ms

5. End-to-End Operational Pipeline Latency Profile

Pipeline Stage	Component Architecture	Runtime (s)	Throughput Share
1. Hydrodynamic Prediction	Pre-trained QIFCP & Physics Engine	0.000 s	0.0%
2. Multi-tier Strategy Optimization	QPSO Composition, Capacity & Speed	0.000 s	0.0%

3. Reliability & Demand Verification	Schedule Buffer & Weather Degradation	0.000 s	0.0%
4. Corridor Deployment Allocation	Vessel Assignment & Port Slotting	0.000 s	0.0%
Total Platform Workflow	End-to-End Complete Execution	0.036 s	100.0%